

Satellite Intelligence from Open Data

Verified data access · six working proof-of-concept businesses · what paid data unlocks · a real-estate deep dive with a JD Fields & Company case study

7 data sources verified live ~10 min freshest imagery 10 m sharpest free 0 API keys 6 POCs built & tested

Executive summary

The most up-to-date imagery we can access for free is about ten minutes old — NOAA's GOES-19 full disk, refreshed every 10 minutes, no key, no contract. Below it on the freshness ladder: hourly NO2 pollution (NASA TEMPO), daily global 250–375 m optical (VIIRS), 10 m optical every ~5 days (Sentinel-2), and 10 m radar every 1–6 days that sees through cloud and darkness (Sentinel-1).

The free ceiling is ~10 m resolution: ships, buildings, fields, plumes and construction are visible; people and cars are not. That line is exactly where the paid market begins — and per-shot pricing now starts around \$300, not six figures (Section 4).

Six businesses were prototyped end-to-end in the repo, each processing real scenes during a single working session. Recommended wedge: dark-vessel detection — free global radar plus free ship-transponder feeds is a real product with proven buyers. For steel-industry buyers like JD Fields, the same pipeline becomes a construction-starts lead engine (Section 6).

1 · What we verified we can access for free

Every row was hit live from this project on 2026-08-11 over anonymous HTTPS — zero credentials, zero cost.

Source	What	Cadence	Latency	Resolution
NOAA GOES-19	Full-disk visible/IR, geostationary	10 min	~10 min	0.5–2 km
NASA TEMPO	NO2 / air quality, North America	hourly	hours	2–5 km
NASA VIIRS/MODIS	Daily global optical, fires, aerosol, nightlights	daily	3–24 h	250–375 m
Sentinel-1 (radar)	C-band SAR — night and through-cloud	1–6 days	hours–1 d	~10–20 m
Sentinel-2 (optical)	13-band multispectral, range-readable COGs	~5 days	hours–2 d	10 m
Sentinel-5P	NO2, CH4, CO, SO2 — global	daily	~1 d	5.5 km
Copernicus Data Space	Backup catalogue, all Sentinels	—	—	—

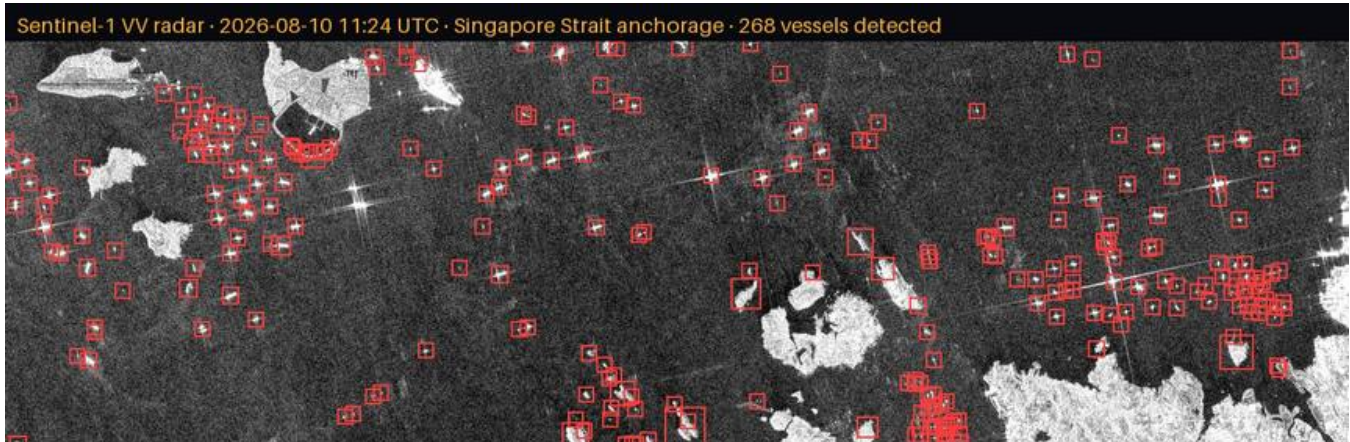
Engineering result that keeps this nearly free to operate: cloud-optimized GeoTIFFs allow windowed HTTP range reads — the ship-detection POC read a 33×19 km window (a few MB) out of a ~1 GB radar scene. Fire detections arrive as vector tiles with exact coordinates.

2 · Six prototyped businesses (all built, all tested)

Each POC has a live page in the app (repo: [hhouston/cheerio_scraper](#), branch [claude/satellite-imagery-business-analysis-pjcaqn](#)) with method, market and unit economics. Highlights, ordered by conviction:

2.1 · Dark Vessel Watch — recommended wedge · public interest

268 vessels detected in one Sentinel-1 radar pass over the Singapore Strait anchorage; hull estimates 40–181 m. Radar detections minus AIS transponder matches = dark ships. Buyers: fisheries regulators, NGOs, marine insurers, commodity desks. Precedents: Global Fishing Watch, Spire, Unseenlabs.

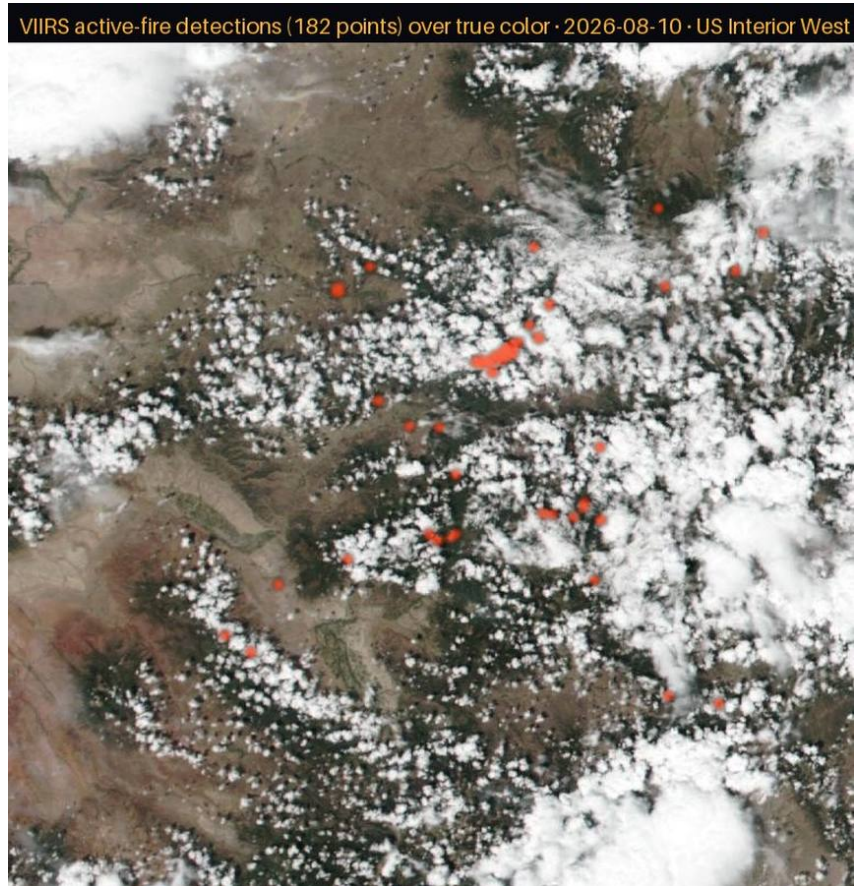


Every red box is a ship — one radar pass, auto water-masked, robust-threshold detection. Radar works at night and through cloud. Sentinel-1 SAR, 2026-08-10 11:24 UTC.

Honest limits: revisit is 1–6 days per site — monitoring, not continuous tracking; the “dark” classification needs the AIS join (a free key away — Section 7).

2.2 · Wildfire & Disaster Watch — public interest

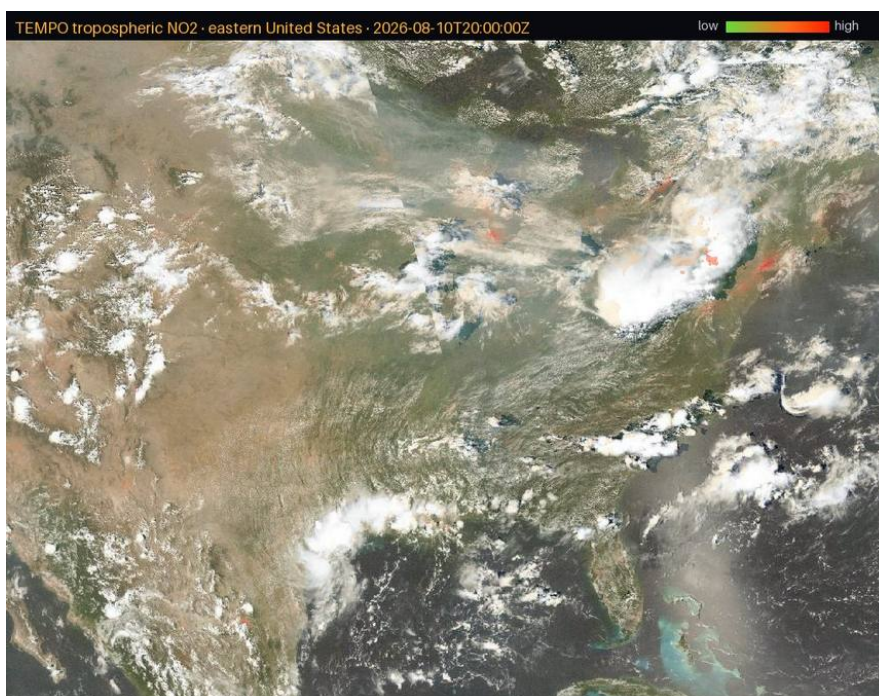
The pipeline scanned **1,696 fire detections** across the western US and auto-located the largest complex (Colorado Rockies) with no human input. Buyers: utilities (PG&E settled \$13.5B in fire liability), insurers, land managers — per-portfolio proximity alerts at \$50k+/yr per utility. Precedents: Pano AI, OroraTech, Watch Duty.



Largest active western-US fire on 2026-08-10, found automatically; dots are 375 m VIIRS thermal detections over same-day true color.

2.3 · Emissions Monitor — public interest

NASA TEMPO measures NO₂ every daylight hour from geostationary orbit — hourly emissions accountability. Sells to ESG platforms, enforcement, litigation, journalism; IRA methane fees (\$900–1,500/ton) make quantified leaks financially material. Precedents: Climate TRACE, Kayrros, GHGSat.



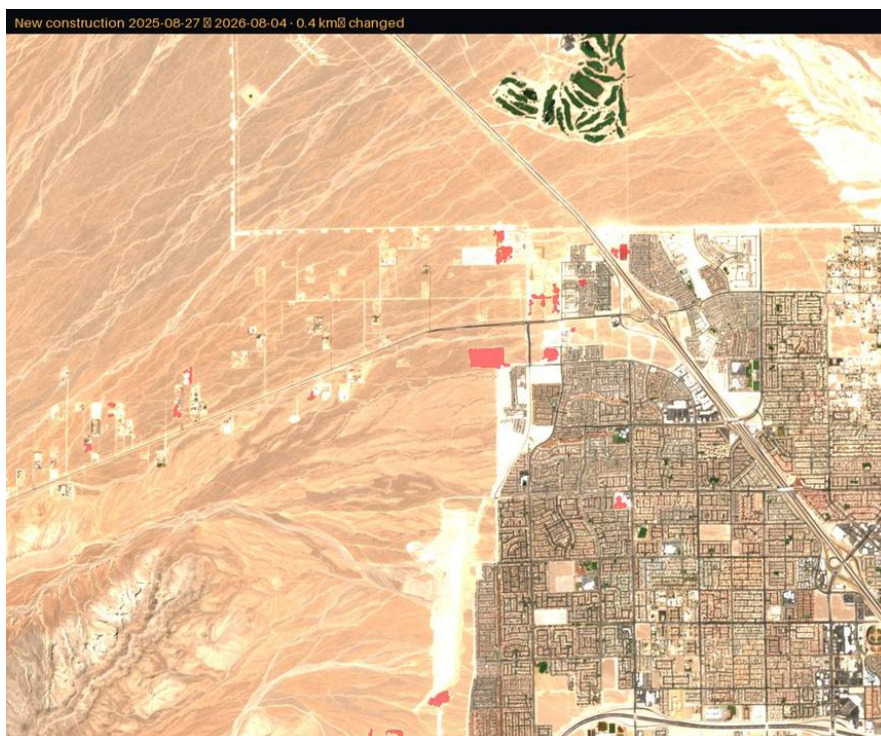
One afternoon hour over the eastern US: the New York plume resolves individually — raw material for facility-level attribution. TEMPO L3, 2026-08-10 20:00 UTC.

2.4 · Route Overwatch — public interest, consent-based

“Have a satellite watch your route”: parse a GPX track its owner shares, pull the freshest 10 m chip at each waypoint, flag active fires along the corridor. \$5–15/mo consumer; \$100+/crew/mo B2B. Honest physics: free imagery gives fresh terrain briefs, not live following — real-time looks are the commercial-tasking premium tier (Section 4). Watching anyone who has not consented is out of scope by design.

2.5 · Construction & Property Intel

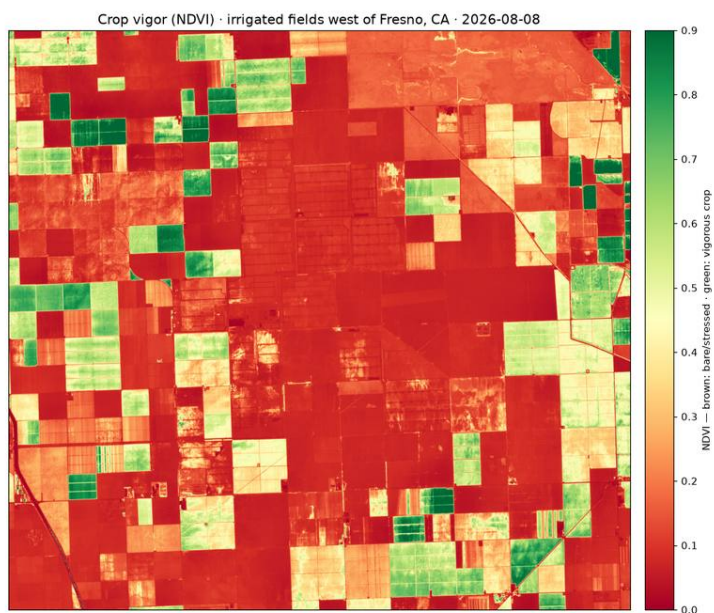
Same grid, same season, one year apart over the NW Las Vegas fringe: **0.4 km²** of new grading and roofs traced along the growth frontier. This POC is the seed of both the real-estate products (Section 5) and the JD Fields case study (Section 6).



Red = desert that brightened the way graded lots and new roofs do, Aug 2025 → Aug 2026. Sentinel-2, season-matched pair.

2.6 · Crop Health Monitor

Field-by-field vigor (NDVI) west of Fresno every ~5 days, plus an April→August growth curve (0.20 → 0.31). Incumbents charge \$1–3/acre/season on free data; the wedge is price and API-first delivery.



Green squares are thriving irrigated fields; brown is fallow or harvested. Sentinel-2, 2026-08-08.

3 · Paying for better satellites: the options ladder

Verified against current (Aug 2026) offerings. The market has changed: per-shot tasking is now self-serve and starts at hundreds of dollars, not six-figure contracts.

Tier	Providers	Cost	Doors it opens
Free public	Sentinel-1/2, GOES, VIIRS, TEMPO, S5P	\$0	Everything the six POCs do today; 10 m ceiling
Free commercial programs	Umbra Open Data (25 cm SAR, ports incl.), Vantor/Maxar disaster imagery, NASA CSDA research access	\$0	Prototype sub-meter products before spending anything
Pay-per-shot marketplaces	SkyFi (also SkyWatch, UP42)	archive from \$25; tasking ~\$300 (50 cm optical, 25 km ²), ~\$812 (30 cm), ~\$675/scene (SAR)	Cars, roofs, ship identification; on-demand looks with no contract
Subscription monitoring	Planet — daily ~3 m of the whole landmass	self-serve from ~\$2,700/yr; 30-day free trial	Cadence: construction and crop change detection go from yearly/weekly to daily
Direct SAR tasking	Umbra, Capella, ICEYE (all hold new NRO contracts)	~\$500–1,500/task via API	Programmatic all-weather, day/night looks; guaranteed revisit
Specialty sensors	Hyperspectral (Pixxel, Tanager, Wyvern); thermal (SatVu, OroraTech); RF (HawkEye 360, Unseenlabs)	enterprise \$\$	Methane per-facility, building heat loss, radio-emission ship geolocation

Recommended spend path: **\$0 first** (Umbra's open SAR already covers ports — feed it into the dark-vessel POC), then **one ~\$300–675 SkyFi experiment** to validate a premium tier, then **Planet's free 30-day trial** — converting to ~\$2.7k/yr only once a construction or crop customer exists.

4 · New business ideas unlocked by non-free APIs

Each idea below is impossible (or toothless) at 10 m but becomes real at the listed tier:

Idea	What you'd sell	Data tier needed
Retail & parking analytics	Store-visit trends per chain/quarter for investors	50 cm optical tasking or Planet SkySat
Oil-tank fill monitoring	Crude inventory estimates from floating-roof shadows	sub-meter optical or SAR; Umbra open data to prototype
Insurance claims triage	Post-storm roof-damage scoring per policy address	30–50 cm optical after events
Construction draw verification	Lender-grade progress verification per loan draw	50 cm tasking, monthly per site
Port & logistics intelligence	Container-dwell and berth-occupancy indices	weekly SAR tasking (weather-proof)

Idea	What you'd sell	Data tier needed
Commodity stockpile tracking	Coal/ore/steel yard volumes as market signals	sub-meter optical + SAR time series
Methane super-emitter alerts	Per-facility leak detection & fee-exposure estimates	hyperspectral partnerships (GHGSat, Carbon Mapper)
Building energy audits	Heat-loss scores per commercial roof, city-scale	high-res thermal (SatVu)
Dark-vessel RF fusion	Geolocate ships by radio emissions even with AIS off	RF sensing (HawkEye 360 / Unseenlabs) partnerships
Real-time asset overwatch	On-demand look at a specific site within hours	direct SAR/optical tasking APIs

5 · Real-estate deep dive

Real estate is the most natural commercial home for this pipeline: the construction POC already produces the core signal (where land is being converted), and every product below is a packaging of that signal for a different buyer.

Growth-frontier maps (free data, works today)

Quarterly metro-fringe change maps — which corridors are being graded, at what pace, versus prior quarters. Buyers: homebuilders, land brokers, building-products distributors. The NW Las Vegas POC is literally this product for one window.

Land acquisition due diligence (free data)

A decade of Sentinel/Landsat history per parcel: flooding scars, vegetation trends, informal dumping, road access appearing/disappearing. Sells as a per-parcel report attached to land deals.

Infrastructure-timing signals (free data)

New roads, utility corridors and interchanges visibly under construction re-price nearby land 12–24 months before completion. Automated alerts per watchlist parcel.

Assessor & appraisal feeds (free data)

County assessors miss new improvements between physical inspections; a change-detection feed flags parcels whose footprint brightened or grew — recovered tax base pays for itself.

Construction-stage classification (Planet tier, ~\$2.7k/yr)

Daily 3 m revisit turns “something changed this year” into “this site was cleared in March, graded in April, foundations in May” — the timing layer sales teams and lenders actually act on.

Lender draw verification (SkyFi tasking, ~\$300/site-check)

Construction lenders release draws against claimed progress; a 50 cm tasked image per draw beats a site visit on cost and beats trust on risk.

Insurance & climate underwriting (mixed tiers)

Wildfire fuel proximity and flood-scar history from free data; roof condition and defensible-space verification from sub-meter tasking.

6 · Case study: JD Fields & Company (jdfields.com)

JD Fields & Company (est. 1985, HQ Houston) is a steel supplier with two divisions: a pipe division (150,000+ tons of carbon steel pipe in inventory — ERW, DSAW and seamless; certified API line pipe grades B through X80, serving oil & gas) and a piling division (H-bearing piles, hot-rolled and cold-formed sheet piling, pipe piling for deep foundations). Notably, the company already tracks the active drilling-rig count as a leading indicator of demand — they are already a satellite-intelligence customer in spirit; they just don't have the satellite yet.

What each existing POC becomes when pointed at JD Fields' business:

Use case	How it works	Which division wins
Piling lead engine	The construction-change POC pointed at Gulf Coast metros and infrastructure corridors: detect clearing, grading and pile-rig mobilization weeks before structures rise. Deep foundations (high-rises, bridges, ports, flood walls) are exactly where piling bids happen — reps call while competitors wait for permit databases.	Piling — lead generation
Ahead-of-rig-count drilling intel	They watch rig counts, a lagging-to-coincident indicator. Sentinel-2 change detection spots new well pads being cleared and pipeline right-of-ways being trenched in the Permian/Eagle Ford before rigs arrive — a leading regional demand forecast for line pipe.	Pipe — demand forecasting
Storm-response targeting	After Gulf hurricanes, seawall, levee and flood-control programs spike sheet-piling demand. GOES (10-min) + VIIRS damage mapping + coastal change detection identify which municipalities and contractors will be buying, first.	Piling — event-driven sales
Yard & port logistics	Monitor their own stockyards' laydown footprint over time and steel cargoes arriving at the Port of Houston. Umbra's free open SAR already covers major ports — a \$0 prototype.	Both — operations
Project-progress delivery timing	Watch supplied jobsites' visible progress to time next deliveries and anticipate reorder points without waiting for the GC to call.	Both — account management
Market-supply intelligence	Import terminal activity and competitor mill-yard inventories as a supply-side signal for pricing steel.	Both — pricing (needs sub-meter tier)

Starter stack for a JD Fields pilot: **\$0** — run the construction POC over Houston metro fringe + the Permian, weekly, and Umbra open-data ports; **~\$1,000** — two SkyFi taskings to validate yard-inventory reads; **~\$2.7k/yr** — Planet monitoring for daily stage classification on the top 50 leads. The pitch to them is one sentence: “you already buy rig-count data — this is rig-count data for everything else you sell.”

7 · AIS: which key to get

Get two free keys today: aisstream.io (free real-time global AIS websocket — the correlation feed for the dark-vessel POC; stream-only, so we run a small recorder for regions we process) and [Global Fishing Watch](#) (free; vessel identity, fishing effort, and their published SAR-detection datasets for benchmarking). Upgrade path: Datalastic (~€99–199/mo self-serve REST with history) → Kpler at enterprise scale — Kpler now owns MarineTraffic, FleetMon and Spire's 100+ satellite AIS constellation, which is what matters mid-ocean where no shore receiver hears. Free historical US-waters AIS for backtesting: NOAA MarineCadastre.

8 · Sequencing

1. Ship Dark Vessel Watch as the flagship: AIS keys → correlation → public demo map (Singapore, Gulf of Guinea, South China Sea).
2. Wildfire asset alerts as first revenue — clearest buyer, near-real-time, free data.
3. Construction/real-estate feeds — including the JD Fields-style pilot, the fastest route to a paying design partner in Houston.
4. Emissions as data-API side product; Route Overwatch as the consumer play once the platform exists.

9 · Licensing, ethics, cost to scale

NASA/NOAA imagery is public domain; Copernicus Sentinel data is free for commercial use with attribution. 10 m pixels cannot resolve individuals — these products monitor activities and places, not people; Route Overwatch reads only tracks their owners share, and nothing here approaches targeting-grade fidelity. Cost to scale: data \$0 until commercial tasking enters the roadmap; compute is cents per region-day; the real costs are AIS licensing at scale and sales.

Working app + pipeline: github.com/hhouston/cheerio_scraper, branch `claude/satellite-imagery-business-analysis-pjcaqn` · run with `pnpm install && pnpm dev` · imagery in this report was produced from live scenes by `tools/eo-pipeline`. Contains modified Copernicus Sentinel data (2026).